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MILITARY SPECIFICATION
FOR
SWITCH, ARM-SAFE
MARK 60 MOD 1

This specification has been approved by the Naval Ordnance Systems Command, Department of the Navy

1. SCOPE

1.1 This specification establishes the requirements of one type of ARM-SAFE Switch Mark 60 Mod 1. (See 6.1 for intended use and description.)

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, form a part of this specification to the extent specified herein, except as noted.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation, Methods of
MIL-E-5272	Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-L-10547	Liners. Case and Sheet, Overwrap, Water-Vaporproof or Waterproof Flexible

FEDERAL

PPP-B-621	Boxes, Wood, Nailed and Lock Corner

FSC - 1135

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STANDARDS

FEDERAL

FED-STD-102	Preservation, Packaging, and Packing Levels

MILITARY

MIL-STD-105D	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-167	Mechanical Vibration of Shipboard Equipment

DRAWINGS

BUREAU OF NAVAL WEAPONS

LD 497919	Switch, Arm-Safe, Mark 60 Mod 1. and all drawings listed thereon
LD 540656	Torque Tester Assembly

PUBLICATIONS

BUREAU OF NAVAL WEAPONS

OD 12975	Operation and Maintenance of ARM/SAFE Switch Torque Tester for ARM/SAFE Switch Mark 60 Mod 1

(Copies of specifications, standards, drawings, and publications requested by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General requirements. Unless otherwise specified in the contract or order, all materials and treatments of materials and parts used in the manufacture of the switch shall be in accordance with the requirements specified herein and with the applicable documents listed in LD 497919.

3.2 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the switch shall be manufactured using the

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methods and procedures proposed for production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract.

3.3 Performance and product characteristics. The switch shall meet the following performance and product characteristics.

3.3.1 Circuit requirements. The following circuit resistance and insulation resistance requirements shall apply after subjecting the wiring harness to repeated bending from the relaxed configuration into the test configuration. (See Figure 1.)

3.3.1.1 Insulation resistance. After being conditioned at a temperature of 75 plus or minus 5 degrees Fahrenheit (F) and a relative humidity of 75 plus or minus 5 percent for a period of not less than twenty-four hours, the insulation resistance shall be not less than 50 megohms. When the insulation resistance is measured during or subsequent to the environmental tests as specified in 4.5, there shall be no temperature or humidity conditioning other than that required by the individual test and the insulation resistance shall then be 20 megohms minimum. The test potential shall be 500 plus or minus 50 volts DC. The insulation resistance shall be measured between all circuits not electrically connected and between all circuits and the case.

3.3.1.2 Circuit resistance. The resistance between the following harness terminals shall be not greater than 0.10 ohm. (See Figure 2.)

SAFE Position Terminals	ARM Position Terminals
52 to 64 52 to 75	49 to 55 65 to 81
51 to 59 52 to 67 52 to 76	50 to 56 71 to 79
52 to 51 52 to 68	51 to 52 82 to 66
52 to 60 52 to 69 53 to 55	57 to 59
52 to 63 52 to 70 53 to 56	58 to 60

3.3.2 Manual actuation torque. The switch shall require a torque of not greater than 40 inch-pounds to rotate the handle from the SAFE to ARM position and from the ARM to the SAFE position. After environmental tests of 4.5 the torque shall not exceed 60 inch-pounds.

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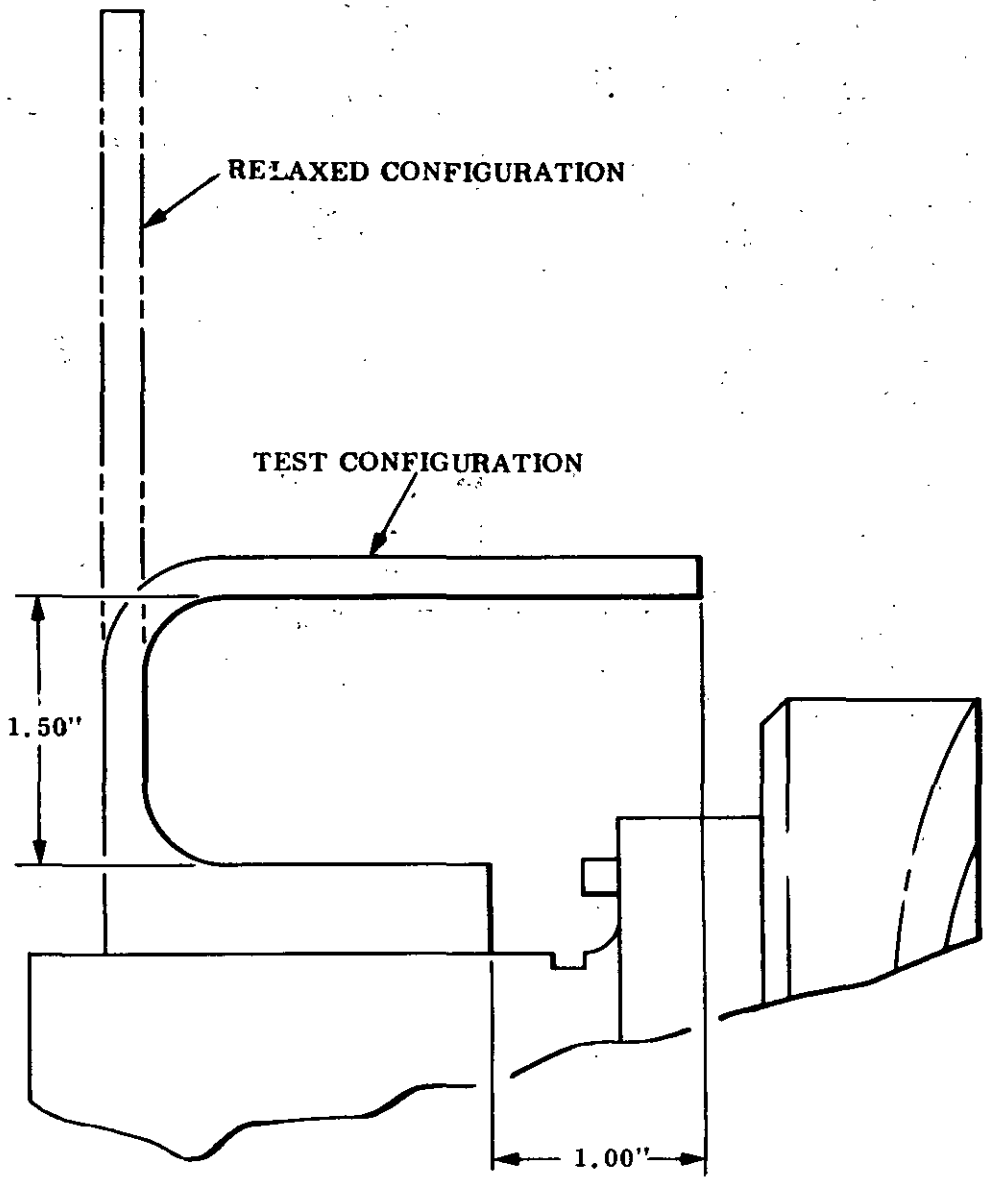


Figure 1. Wiring Harness Circuit Tests Configuration

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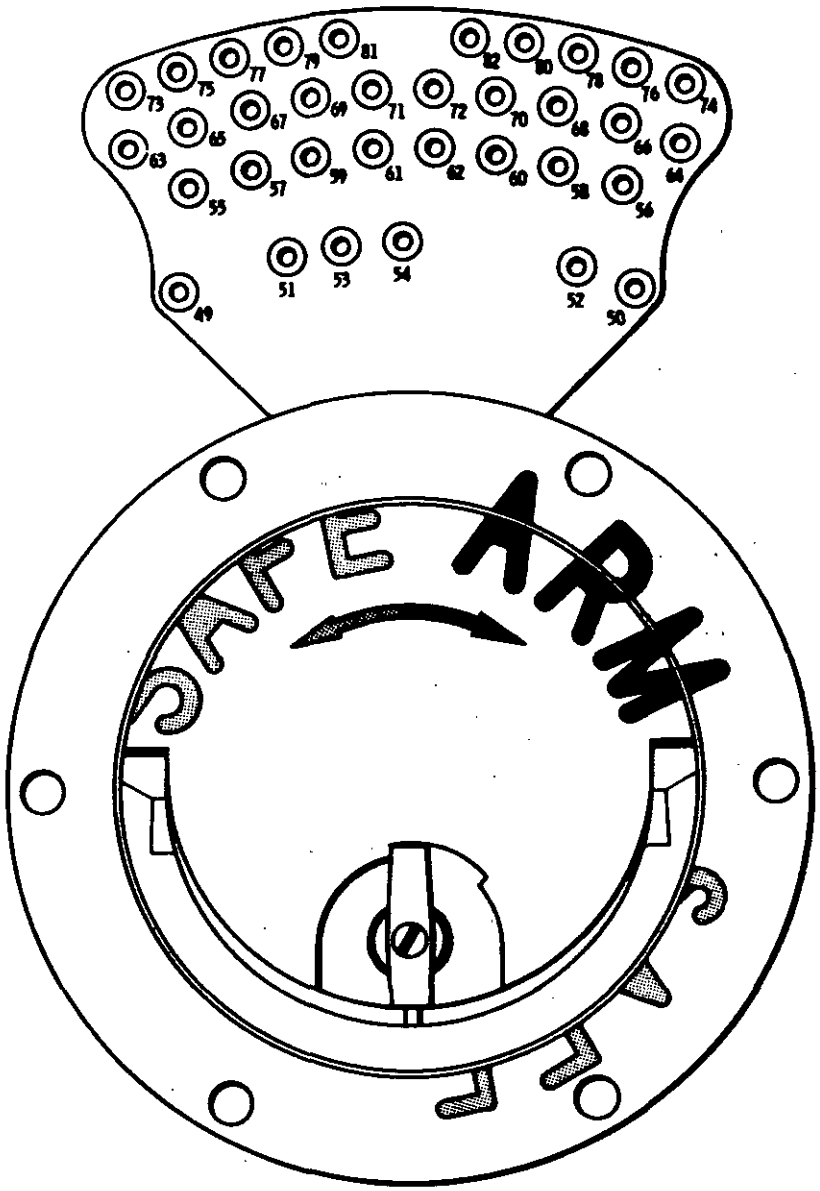


Figure 2. Testing Terminals

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3.4 Environmental requirements. Unless otherwise specified, the switch shall be capable of conforming to the performance and product characteristics of this specification after being exposed to the environments and all conditions specified herein.

3.4.1 Temperature.

3.4.1.1 Operating condition. The switch shall meet the performance and product characteristics while at temperatures from 35 to 120 degrees F.

3.4.1.2 Storage condition. The switch shall withstand temperatures from minus 65 to plus 160 degrees F.

3.4.2 Salt spray. The switch shall withstand salt spray conditions as specified in MIL-E-5272 Procedure I on the exposed surface as shown in Figure 3.

3.4.3 Acceleration. The switch shall withstand an acceleration of 25g (g = unit of gravitational acceleration) for 4 seconds applied in the direction shown in Figure 4.

3.4.4 Vibration.

3.4.4.1 Standby condition. The switch shall withstand vibration as specified for Type I equipment in MIL-STD-167 up to 25 cycles per second (cps).

3.4.4.2 Operating condition. The switch shall withstand vibration through a frequency range of 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less.

3.4.5 Shock.

3.4.5.1 Standby condition. The switch shall withstand shocks of 40g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 5 milliseconds.

3.4.5.2 Operating condition.

3.4.5.2.1 Major. The switch shall withstand a shock of 2400g peak magnitude in the direction shown in Figure 4. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 millisecond. Thereafter, the switch shall meet the requirements for circuit and insulation resistance in the ARM position only.

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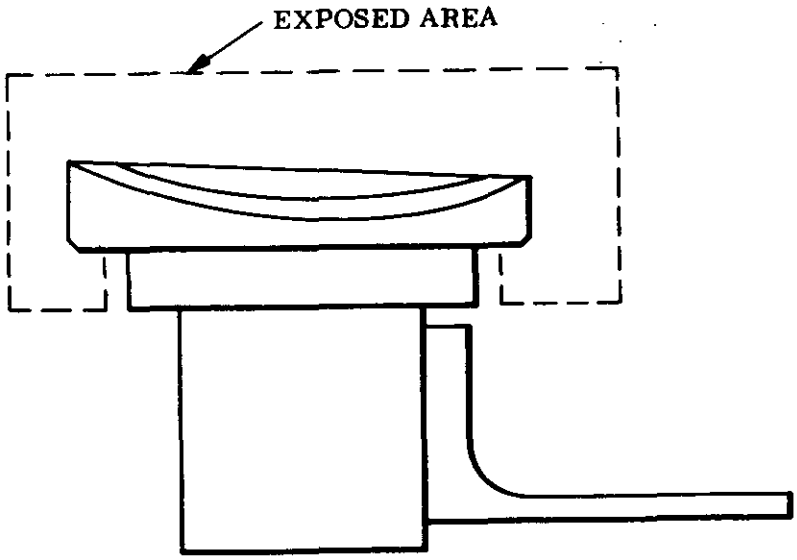


Figure 3. Environmental Exposure Area

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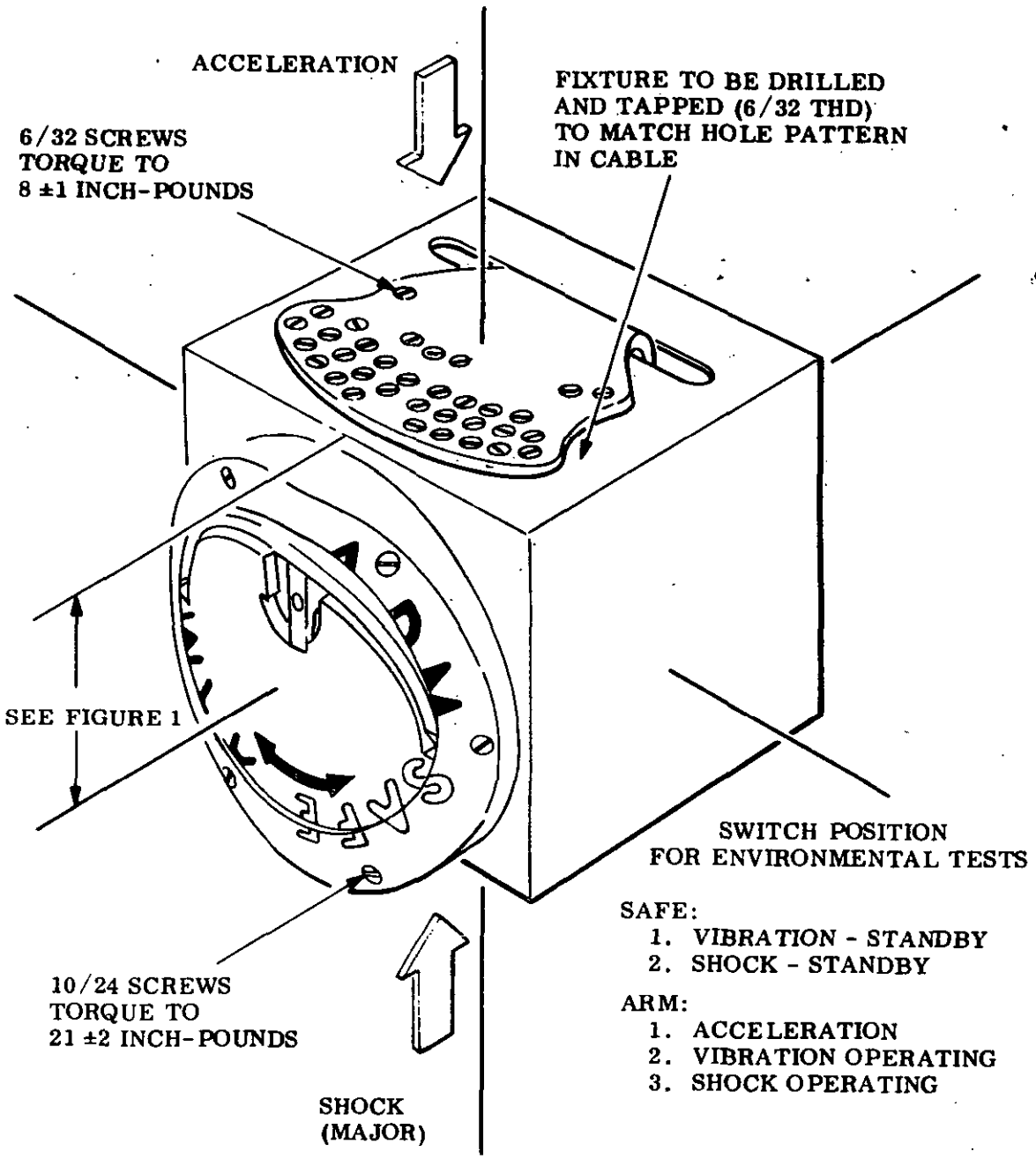


Figure 4. Vibration, Shock, and Acceleration Test Axis and Suggested Mounting Fixture

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3.4.5.2.2 Minor. The switch shall withstand two shocks of 1200g peak magnitude, 90 degrees apart and perpendicular to the major shock axis. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 millisecond. Thereafter, the switch shall meet the requirements for circuit and insulation resistance in the ARM position only.

3.4.6 Pressure. The switch shall withstand a hydrostatic pressure of 750 psig on the exposed area as shown in Figure 3 with no damage or leakage.

3.4.7 Current capacity.

3.4.7.1 Current capacity, 5 amperes. The switch shall be capable of carrying a direct current of 5 amperes through the following circuits:

SAFE Position Terminals	ARM Position Terminals
52 to 75 52 to 63	71 to 79
52 to 64 52 to 69	81 to 65
52 to 70 52 to 68	82 to 66
52 to 76 52 to 67	

3.4.7.2 Current capacity, 10 amperes. The switch shall be capable of carrying a direct current of 10 amperes through the following circuits:

SAFE Position Terminals	ARM Position Terminals
53 to 55	59 to 57 49 to 55
53 to 56	60 to 58 50 to 56

3.4.7.3 Current capacity, 15 amperes. The switch shall be capable of carrying a direct current of 15 amperes through the following circuits:

SAFE Position Terminals	ARM Position Terminals
52 to 60 51 to 59	51 to 52
52 to 51	

3.4.8 Humidity. The switch shall withstand relative humidity up to 95 percent as specified in MIL-E-5272, Procedure I, on the exposed area shown in Figure 3.

3.4.9 Fungus. The switch shall withstand fungus conditions as specified in MIL-E-5272, Procedure I, on the exposed area shown in Figure 3.

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3.4.10 Sand and dust. The switch shall withstand sand and dust tests as specified in MIL-E-5272, Procedure II, on the exposed area shown in Figure 3.

3.5 Workmanship. The workmanship and finish shall be of sufficiently high grade to assure satisfactory operation consistent with the requirements of this specification. All parts used in the assembly shall be thoroughly clean and free of excess materials, chips, and loose or splattered foreign materials. The standards of workmanship exhibited in any approved preproduction sample, subject to any qualification stated in the Government's notice of approval, shall be determinative of the requirements of the contract relative to workmanship insofar as not specifically covered by applicable specifications.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. Inspection of the switch shall be classified as follows:

a. Preproduction and periodic production inspection. (See 4.5.)

b. Quality conformance inspection. (See 4.6.)

4.3 Sampling. (See 6.2.1.)

4.3.1 Lot size. The term "lot" shall mean "inspection lot": that is, a collection of product submitted by a supplier for inspection. Unless otherwise specified in the contract or order the number of product shall be three hundred (300).

4.3.2 Preproduction sample. A preproduction sample of twenty (20) switches, manufactured in accordance with 3.2, shall be subjected to the preproduction tests detailed in 4.5. Acceptance of the preproduction sample shall be based on no defects in the sample. Further production of the switch by the supplier, prior to approval of the preproduction sample, shall be at the supplier's risk.

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4.3.3 Periodic production sample. From each succeeding lot of three hundred (300) or less, a random sample of ten (10) switches shall be selected. These sample units shall be divided into two groups consisting of five (5) units each and subjected to the tests of 4.5 as indicated in Table I. The switches shall be considered satisfactory if they meet all specification and pertinent drawing requirements and withstand the tests specified in 4.5. When a defective unit is found in any lot, acceptance of all units in the lot shall be withheld and the contractor shall:

a. Determine the cause of the defect.

b. Take the necessary action, with the concurrence of the procuring activity, to correct or eliminate the defect from all units in the lot and, if so directed by the procuring activity, from those units previously shipped.

c. Submit to the procuring activity for approval, the proposed corrective action to prevent recurrence of the defect.

4.3.4 Quality conformance inspection sampling. Samples shall be selected as specified in 4.6 and shall be used to determine the acceptability of the switch inspected as specified in 4.6.

Table I. Preproduction and Periodic Production Test Sequence

Test	Paragraph Number	Test Quantities Preproduction	Periodic	
Quality Conformance Inspection	4.6.1	20	5	5
Temperature, Storage Condition	4.5.1.1	20	5	5
Current Capacity	4.5.1.2	20	5	5
Humidity	4.5.1.3	20	5	-
Fungus	4.5.1.4	20	-	-
Quality Conformance Inspection	4.6.1	20	5	-
Sand and Dust	4.5.1.5	20	-	-
Salt Spray	4.5.1.6	20	-	5
Vibration, Standby Condition	4.5.1.7	20	-	5
Shock, Standby Condition	4.5.1.8	20	-	5
Quality Conformance Inspection	4.6.1	20	-	5
Acceleration	4.5.1.9	20	-	5
Vibration, Operating Condition	4.5.1.10	20	5	5

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Table I. Preproduction and Periodic Production Test Sequence (Continued)

Test	Paragraph Number	Test Quantities Preproduction	Periodic	
Shock, Operating Condition	4.5.1.11	20	5	5
Pressure	4.5.1.12	20	-	5
Temperature, Operating Condition	4.5.1.13			
Low Extreme	4.5.1.13.1	20	5	-
Quality Conformance Inspection	4.6.1	20	5	-
High Extreme	4.5.1.13.2	20	-	5
Quality Conformance Inspection	4.6.1	20	-	5

4.4 Test equipment and test conditions.

4.4.1 Quality conformance inspection equipment. The following items of test equipment are required to perform the quality conformance inspection tests:

Equipment	Capability
a. DC power supply, current regulated	Regulate direct current at 100 plus 0 minus 10 milliampere through the test item load (0.10 ohm maximum)
b. Millivoltmeter	Measure 0 to 15 millivolts DC: accuracy plus or minus 3 percent
c. Conditioning chamber	Maintain temperature at plus 75 plus or minus 5 degrees F and a relative humidity of 75 plus or minus 5 percent
d. Megohmmeter	Measure an insulation resistance of 0 to 100 megohms at a potential of 500 volts DC with an accuracy of plus or minus 10 percent at 20 and 50 megohms
e. Torque tester assembly	Conform to LD 540656. Operate in accordance with OD 12975

4.4.2 Preproduction and periodic production test equipment. The following items of test equipment, in addition to selected items of quality conformance inspection equipment and pieces of equipment recommended by applicable specifications or standards referenced within test procedure paragraphs, are required to perform the preproduction and periodic production tests.

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Equipment	Capability
a. Temperature chamber	Maintain temperature at: Minus 65 plus 5 minus 0°F Plus 35 plus 5 minus 0°F Plus 120 plus 0 minus 5°F Plus 160 plus 0 minus 5°F
b. Vibration testing machine	Control frequency from 20 cps to 2000 cps while subjecting the test item to 0.10-inch double amplitude or 10g acceleration, whichever is less: maintain direction of vibration along specified axes
c. Centrifuge	Maintain acceleration at 25 plus 2.5 minus 0g
d. Shock equipment, standby	Apply shocks of 40 plus or minus 4g: a one-half sine wave pulse shape having a duration of 5 plus 3 minus 0 milliseconds: maintain direction along specified axes
e. Shock equipment, operating	Apply shocks of up to 2400g plus or minus 10 percent; a one-half sine wave pulse shape having a duration of 1 plus or minus 0.25 millisecond: maintain direction along specified axes
f. Pressure chamber	Provide a hydrostatic pressure of not less than 750 psig plus or minus 25 psig for 30 plus or minus 5 seconds
g. DC power supply	Regulate direct current at 5 plus 0 minus 0.5, 10 plus 0 minus 1 and 15 plus 0 minus 2 amperes through test item load (0.10 ohm max.)
h. Voltmeter	Measure 0 to 2.5 volts DC: accuracy within 2 percent

4.4.3 Test conditions. Unless otherwise specified, the switch shall be subjected to the preproduction and periodic production tests and quality conformance inspection tests under the following conditions:

a. Temperature	Room ambient (65 to 95 degrees F)

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b.	Altitude	Normal ground atmospheric pressure
c.	Humidity (Relative)	Room ambient up to 95 percent

4.5 Preproduction and periodic production inspections. The preproduction and periodic production samples, after satisfactorily passing the quality conformance inspection detailed in 4.6, shall be subjected to the tests specified in 4.5.1.

4.5.1 Test procedure. Subject the switch to the test sequence listed in Table I. Unless otherwise specified in the following individual test paragraph, the quality conformance inspection specified in Table I shall be performed subsequent to environmental tests under test conditions specified in 4.4.3. Failure of the switch to pass any one of these tests shall constitute a defective unit.

4.5.1.1 Temperature, storage condition. The switch, in the SAFE position, shall be subjected to a temperature of minus 65 plus 5 minus 0 degrees F for not less than 48 hours. The temperature shall be raised to plus 160 plus 0 minus 5 degrees F in not less than 2 hours and maintained at this temperature for not less than 48 hours. The unit shall then be subjected to room temperature for not less than 4 hours prior to performing subsequent tests.

4.5.1.2 Current capacity test.

4.5.1.2.1 Five ampere circuits. Connect the voltmeter between the terminals listed and, using the power supply, apply a direct current of 5 plus 0.0 minus 0.5 amperes between the terminals for 90 seconds. The voltage drop shall be not greater than 0.5 volt. (See Figure 2 for terminal location.)

SAFE Position Terminals	ARM Position Terminals
52 to 75 52 to 63 52 to 64 52 to 69 52 to 70 52 to 68 52 to 76 52 to 67	71 to 79 81 to 65 82 to 66

4.5.1.2.2 Ten ampere circuits. Connect the voltmeter between the terminals listed and, using the power supply, apply a direct current of 10 plus 0.0 minus 1.0 amperes between the terminals for 90 seconds. The voltage drop shall be not greater than 1.0 volt. (See Figure 2 for terminal location.)

SAFE Position Terminals	ARM Position Terminals
53 to 55 53 to 56	59 to 57 49 to 55 60 to 58 50 to 56

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4.5.1.2.3 Fifteen ampere circuits. Connect the voltmeter between the terminals listed and, using the power supply, apply a direct current of 15 plus 0.0 minus 2.0 amperes between the terminals for 90 seconds. The voltage drop shall be not greater than 1.5 volts. (See Figure 2 for terminal location.)

SAFE Position Terminals	ARM Position Terminals
52 to 60 51 to 59 52 to 51	51 to 52

4.5.1.3 Humidity. The switch, in the SAFE position, shall be subjected to Humidity Test, Procedure I, as specified in MIL-E-5272, on the exposed area shown in Figure 3.

4.5.1.4 Fungus. Unless otherwise specified in the contract or order, the test shall be performed on preproduction sample units only. The switch, in the SAFE position, shall be subjected to Fungus Tests, Procedure I, as specified in MIL-E-5272 on the exposed area shown in Figure 3.

4.5.1.5 Sand and dust. Unless otherwise specified in the contract or order, this test shall be performed on preproduction sample units only. The switch, in the SAFE position, shall be subjected to Sand and Dust Tests, Procedure II, as specified in MIL-E-5272 on the exposed area shown in Figure 3.

4.5.1.6 Salt spray. Subject the switch, in the SAFE position, to Salt Spray Tests as specified in MIL-E-5272, Procedure I, on the exposed area shown in Figure 3 for a period of 50 hours.

4.5.1.7 Vibration, standby condition. The switch, in the SAFE position, and rigidly mounted as shown in Figure 4, shall be subjected to the vibration tests for Type I equipment as specified in MIL-STD-167 up to 25 cps in each of the three mutually perpendicular axes shown in Figure 4.

4.5.1.8 Shock, standby condition. The switch, in the SAFE position, and rigidly mounted as shown in Figure 4, shall be subjected to shocks of 40 plus or minus 4g peak magnitude with a pulse shape approximating a one-half sine wave, having a duration of 5 plus 3 minus 0 milliseconds. Shocks shall be applied three times in each direction along each of three mutually perpendicular axes shown in Figure 4.

4.5.1.9 Acceleration. The switch, in the ARM position, and rigidly mounted as shown in Figure 4, shall be subjected to an acceleration of 25 plus 2.5 minus 0g in the direction shown in Figure 4, for not less than 4 seconds.

4.5.1.10 Vibration, operating condition. The switch, in the ARM position, and rigidly mounted as shown in Figure 4, shall be subjected to vibration from 20 to 2000 cps at 0.10-inch double amplitude or 10g, whichever is less,

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along each of the three mutually perpendicular axes: The frequency shall be traversed in not less than 2.5 minutes, once in each axis. Thereafter, the switch shall meet the requirements for circuit resistance 4.6.1.2 and insulation resistance 4.6.1.1 in the ARM position only.

NOTE: Instrumentation used for the shock test shall not employ a filter with a cut-off frequency below 10 kc.

4.5.1.11 Shock, operating condition. The switch, in the ARM position, shall be mounted as shown in Figure 4, and subjected to the following tests:

4.5.1.11.1 Major. Subject the switch to a shock of 2400 plus or minus 240g peak magnitude. The shock shall have a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond. Apply the shock in the direction shown in Figure 4. Thereafter, the switch shall pass the tests of insulation resistance 4.6.1.1 and circuit resistance 4.6.1.2, in the ARM position only.

4.5.1.11.2 Minor. Subject the switch to two shocks of 1200 plus or minus 120g peak magnitude. The shocks shall have a pulse shape approximating a one-half sine wave having a duration of 1 plus or minus 0.25 millisecond. Apply the shocks in two directions 90 degrees apart and perpendicular to the major shock axis. Thereafter, the switch shall pass the tests of insulation resistance 4.6.1.1 and circuit resistance 4.6.1.2, in the ARM position only.

4.5.1.12 Pressure. Subject the switch, in the ARM position, to a hydrostatic pressure of 750 plus or minus 25 pounds per square inch applied to the exposed area as shown in Figure 3 for a duration of 30 plus or minus 5 seconds. The switch shall maintain continuity of the closed circuits during pressure and shall not leak.

4.5.1.13 Temperature, operating condition.

4.5.1.13.1 Low extreme. The switch selected from the sample units as indicated in Table I shall be subjected to a temperature of 35 plus 5 minus 0 degrees F for not less than 2 hours and while at that temperature shall pass the quality conformance tests.

4.5.1.13.2 High extreme. The switch selected from the sample units as indicated in Table I shall be subjected to a temperature of 120 plus 0 minus 5 degrees F for not less than 2 hours and while at that temperature shall pass the quality conformance tests.

4.6 Quality conformance inspection. Samples selected using the plans and procedures set forth in Table II of this specification, shall be subjected to the tests and examinations described in 4.6.1.

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Table II. Quality Conformance Tests and Examinations and AQL

Test	Sampling Plan	AQL	Test Paragraph
Insulation resistance test	*	1.00	4.6.1.1
Circuit resistance test			
ARM position	*	0.25	4.6.1.2
SAFE position	*	0.25	4.6.1.2
Manual actuation torque test	*	1.00	4.6.1.3
Visual examination	**	**	4.6.1.4
Preservation, packaging, packing, and marking	**	**	4.7

*MIL-STD-105D Inspection Level II
**One hundred percent inspection: defect is cause for rejection of individual unit.

4.6.1 Test Procedure. The switch shall be subjected to the tests and examinations as outlined herein. Unless otherwise specified, inability of the switch to pass the tests and examinations shall constitute a failure.

4.6.1.1 Insulation resistance. Place the switch in a conditioning chamber and maintain the temperature at 75 plus or minus 5 degrees F and the relative humidity at 75 plus or minus 5 percent for a period of not less than twenty-four hours. Remove the switch from the chamber and within 10 minutes measure the insulation resistance between the mutually insulated terminals listed below. The test potential shall be 500 plus or minus 50 volts DC. The insulation resistance shall be not less than 50 megohms. When this test is performed during or subsequent to the environmental tests specified in 4.5.1, there shall be no temperature or humidity conditioning other than that required by the individual environmental test; the insulation resistance shall then be 20 megohms minimum.

SAFE Position Terminals	ARM Position Terminals
50 82 71 79 66 58 57 65	50 60 82 71 76 68 64 52 53
81 49 56 52 Case	55 57 65 63 67 75 69 70 Case

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4.6.1.2 Circuit resistance test. Apply a direct current of 100 plus 0 minus 10 milliamperes between the following harness terminals using the DC power supply, specified in 4.4.1a. Measure the voltage drop between the terminals using the millivoltmeter specified in 4.4.1b. The voltage drop shall be not greater than 10 millivolts.

SAFE Position Terminals	ARM Position Terminals
52 to 64 52 to 75 51 to 59	49 to 55 65 to 81
52 to 67 52 to 76 52 to 51	50 to 56 71 to 79
52 to 68 52 to 60 52 to 69	51 to 52 82 to 66
53 to 55 52 to 63 52 to 70	57 to 59 58 to 60
53 to 56	

4.6.1.3 Manual actuation torque test. Rotate the switch handle from the SAFE to ARM position and from the ARM to the SAFE position. Measure the torque, required to rotate the handle, using the torque tester assembly specified in 4.4.1e. The torque shall be not greater than 40 inch-pounds. When the test is performed subsequent to the environmental tests of 4.5, the torque shall not exceed 60 inch-pounds.

4.6.1.4 Visual examination. Visual examination shall be made on each switch submitted for delivery to the procuring agency. The switch shall be examined for conformance to the requirements of good workmanship in 3.5. Defect is cause for rejection of individual unit.

4.7 Preservation, packaging, packing, and marking. The switch shall be examined for conformance to the prescribed preservation, packaging, packing, and marking requirements of Section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A or C as specified. (See 6.2b and 6.4.)

5.1.1 Level A.

5.1.1.1 Cleaning. The switch shall be cleaned in accordance with Method C-1, MIL-P-116.

5.1.1.2 Drying. The switch shall be dried in accordance with Method D-1, MIL-P-116.

5.1.1.3 Unit packaging. The switch shall be individually packaged in accordance with Method 1A-8, MIL-P-116.

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5.1.1.4 Intermediate packaging. Intermediate packaging of the switch shall be in accordance with the applicable drawings or as specified in the procurement document.

5.1.2 Level C. Cleaning, drying, preservation and packaging shall be in accordance with the supplier's commercial practice, when such practices have been approved by the procuring activity.

5.2 Packing. Packing shall be Level A or C as specified. (See 6.2c and 6.4.)

5.2.1 Level A. Unless otherwise specified in the contract, order or requisition, the switches shall be packed in exterior-type shipping containers, PPP-B-621, of uniform size and shape, with the minimum cube and tare consistent with the protection required. The gross weight of each pack shall be limited to approximately 100 lbs. Containers shall be provided with a case liner, MIL-L-10547, and shall be closed and strapped in accordance with PPP-B-621.

5.2.2 Level C. Switches packaged as specified in 5.1.2 shall be packed to afford protection against damage during direct shipment from the supply source to the first receiving activity for immediate use. Containers shall comply with common carrier regulations applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES

6.1 Intended use and description.

6.1.1 Intended use. The switch is intended for use as a safety device by providing switching for multiple circuits in the fuzing system of a depth charge.

6.1.2 Description. The switch is a device designed to provide manual arming and safing of the fuzing system of a depth charge.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification

b. Required level of preservation and packaging

c. Required level of packing

d. Any exceptions to this specification

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6.2.1 Additional samples. Additional samples required because of failure of any sample(s) to pass the prescribed inspections shall be provided and inspected at the expense of the supplier.

6.3 Definitions. For the purpose of this specification the following definitions have been established.

6.3.1 Unit condition.

6.3.1.1 Operating. The condition of the switch during missile operation.

6.3.1.2 Standby. Standby is the condition of the switch when aboard ship and is assembled into the end item.

6.4 Preparation for delivery criteria. Criteria for use of the proper level of preservation, packaging, and packing shall be as specified in FED-STD-102.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION
MIL-S-82520(OS) SWITCH, ARM-SAFE MARK 60 MOD 1

ORGANIZATION *(Of submitter)*

CITY AND STATE

CONTRACT NO.

QUANTITY OF ITEMS PROCURED

DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*

DATE

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